Measurement-Dependent Clustering in the Growing Urban Observatory Sensor Network

Ammar Arbaaeen[2], **Wael Alnahari**[1], **Ammar Alsaig**[2], **Tawfiq Hasanin**[1], **and Wasem Alnahri**[3]

[1]**Department of Information Systems, Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah 21589, Saudi Arabia**
[2]**Department of Computer Science, College of Computer and Information Systems, Umm Al-Qura University, Mecca 24381, Saudi Arabia**
[3]**Independent Researcher, Mecca 24331, Saudi Arabia**

Corresponding author:
Wael Alnahari[1]

Email address: walnahari0004@stu.kau.edu.sa

Postal address: MDFA3058, 3058 Prince Nasser bin Abdulaziz Street, King Fahd District, Makkah 24342, Saudi Arabia

ABSTRACT

Urban sensing studies describe a network from a single slice, so they cannot show whether its structure changes as it grows. We census the Newcastle Urban Observatory instrument population annually, 2013–2026 (41 to 706 identifiers), and ask how much of the apparent strengthening of nearest-neighbour clustering is a property of the deployment, not the measurement. The Clark–Evans index falls 67% on each year's own bounding box and identifiers, 57% on a fixed reference hull, and 45% once identifiers at one coordinate are counted once; measuring each year on its own hull gives 64.7%. Merging registrations within 25 m leaves 4.3%, shrinking monotonically with no plateau, so the tolerance is a choice, not a threshold the data identify. The two controls interact rather than partition the decline: registration removes it on the fixed hull, support matching restores 38.6%, so we report a range, not a decomposition. On each year's own footprint the multiscale decline is -99 m per year, indistinguishable from zero, and rarefaction also removes the trend. Clustering is defined against a homogeneous Poisson process, invariant under translation and rotation of the window; window, density and registration each move the index from that reference value without moving an instrument.

INTRODUCTION

An urban sensor network is usually presented as a fixed capability: so many instruments, covering so much of a city, measuring so many variables. Work on the Internet of Things (IoT) describes city-scale deployments in precisely these terms (Kumar et al.; Jamshed et al.; Zeng et al.; Xu et al.), and the urban observatory literature characterises platforms the same way (Hamed et al.; Yu; Davern et al.). Reviews of sensing infrastructure treat deployment as a design decision taken once and then inherited (Osamy et al.). The published work on the Newcastle Urban Observatory follows this convention. One study forecasts pedestrian flow from 110 sensors (Morris-Wiltshire et al.); another uses 13 instruments on two shopping streets (Komar and James); a third measures air quality at 12 schools (Keast et al.); the equity literature analyses air-quality locations as a placement problem (Robinson and Franklin; Robinson et al.; Zhai et al.; Lin et al.; Zied Abozied et al.). Each is a legitimate design, and each shares one limitation that is rarely stated: the network is treated as whatever the chosen slice contains, at whatever moment the study was conducted.

That is a problem for a longitudinal platform. The observatory has existed since 2013. If its instrument population changed in size, composition, or spatial arrangement over those years, then results anchored to a single slice describe a configuration that may not have existed before or since. Structure is not a nuisance variable here; it is what determines which places are observed and therefore what any statistic computed from the network can mean.

Answering the question requires a spatial index, and the standard one is hazardous in exactly this setting. The Clark–Evans index divides the observed mean nearest-neighbour distance by its expectation under complete spatial randomness, so it depends on three things that all change as a network grows: the reference area, the number of points, and the rule that decides which records count as separate points. A network that doubles its instruments, or registers each device under several identifiers, will move the index with no change at all in where instruments stand. We therefore treat the observed change in the index as a quantity to be attributed to the measurement protocol rather than as a finding to be reported, and the paper's main result is how far that attribution can be pushed.

The reference state for clustering, a homogeneous Poisson process, is invariant under translation and rotation of the observation window. Every departure measured below is a departure from that symmetry, and each of the three factors above can create or erase such a departure without any change in the deployment. We use no symmetry group, and no group-theoretic structure is assumed anywhere in this paper: the sense in which the subject is symmetry is the invariance of the reference state, against which every quantity below is read as a departure. That is the sense in which this paper is about symmetry, and it follows the view of **(author?)** (Xia et al.) that measurement-dependent asymmetry is conditioned on the reference state, the sample and the scale. We do not claim that a city's sensor network resembles its reference state; for a planned deployment that is not in question.

The record also supports one symmetry test directly, and it is reported here rather than left to the methods. The orientations of pairs within 1 km of one another are not uniform (the best-to-worst sector ratio is 1.42), so the pattern has directional structure. The calibrated test is a rotation null: rotating the observed pattern about its centroid inside the fixed window, which preserves the clustering and the pair count, gives a ratio of 1.37 (sd 0.06), so the observed 1.42 is well inside what a rotated copy of the pattern itself produces (1 of eight sectors outside). The pattern's orientation relative to its frame is therefore unremarkable, and we report the test in that form rather than as a detected asymmetry. Note what the test does *not* say: the elongated window does not explain the imbalance either, because a dense complete-spatial-randomness sample in the same hull gives a ratio of 1.01. Both attributions are therefore excluded, and what remains is a directional non-uniformity whose origin this record cannot identify. That test, together with the interaction of the window, the sample size and the registration rule on the clustering index, is what this paper treats as its symmetry content: in both cases an apparent asymmetry is conditioned on the reference state against which it is measured, and the point of the analysis is to say which.

This paper therefore asks: **of the apparent strengthening of clustering in this network, how much survives once the window, the sample size and the registration rule are controlled?** Answering it needs the whole archive and not a slice. The instrument population of an urban observatory has not previously been censused year by year across its life, as far as we can establish: the studies above fix a slice, so the population itself has been assumed rather than measured. The relation to existing work sets out how that census relates to the equity and placement literature; no causal claim is made about why the network was deployed as it was.

Contribution

1. A **per-year census** of the instrument population represented in the archive, 2013–2026, built from every archived file and not from a sample (884 files; the methods).

2. A **factorial comparison** of the change in the Clark–Evans index under the window, density and registration controls, which shows that they interact rather than partition the change. On the fixed hull the mean nearest-neighbour distance scales with population as $N^{-0.603}$ (se 0.068), against the $-\frac{1}{2}$ that densification at constant shape forces, so the densest years are more clustered than density alone predicts but not significantly so ($t = -1.52$). The apparent trend in the index is the sum of that distance term, a count term and a window term, and the identity between them holds to 10^{-15} (the clustering-trend analysis).

3. A **registration-resolution curve** showing how the headline depends on the merge radius. Fixing the window and keeping the identifier unit takes the fall from 67% to 57%; deduplicating to the location unit takes it on to 45% on the same fixed hull. From there the merge radius drives it to 4.3% at 25 m, monotonically and with no plateau, and it reverses sign by 50 m; the nearest-neighbour distribution has no trough to place a threshold in (Fig. 1). The consequence applies beyond this network:

because no feature of the point pattern identifies a radius, the merge is a *bounding parameter* and not one the data estimate, so a deduplicated spatial result is a bound unless the radius is shown to be identified — which, for a smooth nearest-neighbour distribution, it cannot be. The radius is treated that way throughout, and the curve is reported instead of a chosen value. The curve has no plateau for a reason that is identified and not merely unexplained: the co-location is duplicated registration concentrated in one of two published coordinate resolutions, and rounding the other resolution to the same quantum leaves it unchanged, so the artefact is neither a rounding effect nor a free choice of radius (the co-location analysis).

4. A **multiscale analysis on support-matched windows**. The multiscale decline measured on the fixed hull over the published archive — the Besag L function at the 1 km scale, $L(1\,\mathrm{km})$ — is largely a support mismatch: measured on each year's own footprint the same quantity falls to -99 m per year and is not distinguishable from zero, while the function still exceeds an inhomogeneous null preserving each year's intensity at 250 m to 1.5 km in all fourteen years. **The two indices move in opposite senses and the sign of one is not the sign of the other:** a falling R means *stronger* short-range clustering, whereas a falling $L(1\,\mathrm{km})$ means *weaker* clustering at that scale, because L measures excess neighbours within the radius. What survives here is a level difference, not a trend (the multiscale analysis, the clustering-trend analysis).

5. A **membership replacement rate**: the share of annual arrivals and departures that swap places instead of adding to the network is 0.58 at the median, and 562 of the 1899 identifiers leave and return (the population-turnover analysis). This is the one result here whose value does not depend on which of the two units is used, though the frame still sets it.

6. A **negative result and an unresolved one**, established and not merely asserted. The clustering index shows no trend once every year is rarefied to a common sample size. For the clustering index a minimum detectable effect of 0.008 per year at 80 % power shows the design would have detected the change at the scales the rarefaction can see, which are above about 5 km, so the null is informative rather than underpowered. The Gini of occupied cells behaves in the opposite way and is *not* null: uncorrected it rises at +0.013 per year, and rarefied it falls at between 0.004 and 0.011 per year depending on the cell size, with a rank interval that excludes zero in each case. Rarefaction therefore reverses the sign of the concentration trend rather than removing it, so that series is unresolved in either direction (the clustering-trend analysis, the unit-and-concentration analysis).

7. A **provenance audit** of a distributed benchmark's frozen snapshot against the reconstructed population, with the theme composition of its omissions tested as a 2×2 contrast against the identifiers the snapshot holds, not against a base rate that includes the omitted; the audit is in the Supplemental Information (S1).

Relationship to existing work

We state this precisely, because a claim of first-ness would be false. **(author?)** (Xia et al.) argue that apparent asymmetry in urban data is partly a *measurement* property conditioned by reference state, sample and scale, and recommend multiscale reporting. Robinson and Franklin (**(year?)**) integrate observatory locations with socio-economic data to map sensor deserts, and **(author?)** (Robinson et al.) extend that to spatial optimisation of air-quality placement. Neither examines the network as a population evolving over time: Xia et al.'s object is inventory construction in carbon accounting, and the Robinson studies' object is placement equity at a single moment. Our object is instead the geometry of the deployed population across its whole life. Those studies complement this one; they do not pre-empt it. Alternatives to the indices we use are well established in urban spatial analysis (Páez and Scott). Moran's I (Moran) and the Getis–Ord G_i^* family (Getis and Ord) would answer a neighbouring question, which places in the city are locally hot, and are not taken here for a specific reason: both require a spatial weight matrix that must itself be chosen and declared, and our object is the geometry of the whole deployment, not a local map. The multiscale K and L functions in the clustering-trend analysis serve the same descriptive purpose with one declared parameter, the distance r, instead of a weight. These two are cited to record that the alternative was considered and why it was not taken, not as support for a claim.

The observatory has also been characterised as an engineering and governance object (26; 59), and the platform type itself has been treated as a governance object and not only an instrument (Barns; Cape). It is one instance of an IoT deployment in public infrastructure, and the challenges it faces are the general ones: intermittent connectivity, heterogeneous devices and undocumented changes to what a feed reports (Jamshed et al.; Zeng et al.). It has been compared with peer platforms elsewhere (Derbali et al.; Fernandes Barata et al.; Li et al.; Longley). Its operational behaviour is documented in several data-facing studies. **(author?)** (Morris-Wiltshire et al.) show that the structure of missing data governs forecast error across 110 pedestrian sensors. **(author?)** (Chen et al.) derive mobility statistics from traffic cameras. **(author?)** (Altarrazi et al.) build a fault taxonomy evaluated on observatory air-quality data. **(author?)** (Farhadi et al.) use observatory data to evaluate a transport policy intervention, and Wolf et al. (58) combine observatory feeds with traffic-control data for rainfall impact. **(author?)** (Puussaar et al.) open the platform to citizen commissioning, and **(author?)** (Keast et al.) measure school-level air quality. **(author?)** (Rusli et al.) survey urban observatory analytics internationally. Every one of these takes the network as given at the time of study; none asks how the deployed population itself changed.

Two further literatures set the context without addressing the question. Work on network placement asks where instruments *should* go, and treats the existing set as a constraint (Ayati et al.; Sheeja Rani et al.; Nazeeya Anjum et al.; Xu et al.; Malaker and Meng); work on urban analytics consumes whatever the platform emits (Theodorakopoulos and Theodoropoulou; Avram; Yu and Fang; Huang et al.; Enlund et al.). Both are downstream of the deployment process we are trying to observe. The city itself has been studied extensively (Barke; Wood), which is exactly why an instrumented record of it is worth characterising, not assuming.

MATERIALS AND METHODS

Data and the census definition

The observatory publishes monthly comma-separated-value (CSV) bundles through an archive application-programming interface (API); the collection is archived with its own DOI and released under CC BY 4.0 (27). We hold the complete local archive: 884 files covering 2013–2026, about 73.8 GB, holding 3 622 852 820 readings in total. An *archive bundle* here is one published file: a bundle is not one file per month, and the count is not fourteen years $\times$ twelve months. A file holds one theme for one month, so a month with several active themes publishes several files, and the file count per year rises and falls with both the number of themes and the publishing regime. The counts run from 11 files in 2013 to a peak of 75 in 2018, back to 41 in 2025 and 257 in the partial 2026, and the 884 is their sum. Where the file inventory itself moves, as it does between 2017 and 2018, we treat that as a platform-side change and not as a change in the city (the theme-discontinuity analysis). The 2026 slice is partial by construction: the archive held nine months, and 2026 values are marked as partial wherever they are compared with a complete year. Each record carries twelve fields, of which the instrument identity and location are `Sensor Name`, `Raw ID`, and the two centroid coordinates.

An instrument is counted as **present in year** y if it appears in at least one archive file for y. This is an observation-based definition, not a vendor deployment date, and we adopt it deliberately because deployment metadata for individual instruments is not published. Two consequences follow, and Table 1 records them as interpretive boundaries: a silent instrument is indistinguishable from a failed one, and a single reading in a year is sufficient for presence. We therefore report births and last appearances separately (the population-turnover analysis) instead of a net count, which would hide the churn.

Two parsing hazards were identified and controlled. Vehicle files contain unquoted commas that displace columns under a naive split on `,`; all parsing uses a proper CSV reader. The most recent month is not yet archived, because the observatory publishes a month only after it closes. September 2026 was therefore obtained from the live query API and converted to the archive schema. That conversion synthesises the `Units` column from the observatory's own variable catalogue and removes the API-only `Flagged` field. The analysis is therefore cut off at 30 September 2026 and September 2026 is a partial month.

Spatial unit and reference populations

Instrument identity and location differ in this archive, and three counts are easily confused, so we separate them. Over the whole record 1906 distinct identifiers appear. Seven carry coordinates that put them more than 60 km from the projection origin, including two whose positions are not real (they decode to roughly

14000 km away), so they are removed by that cut, leaving 1899 identifiers in the in-range population and dropping seven of the 1906. Those 1899 identifiers occupy 1567 distinct coordinates, so:

- 599 identifiers, or 31.5%, sit at a coordinate that at least one other identifier also occupies (599/1899). This is the number of identifiers that *share* a coordinate.
- 1899 minus 1567 equals 332: the number of identifiers in excess of the number of coordinates, or 17.5% of the population (332/1899). The two are easy to confuse, and they are not the same: the share counts identifiers that are co-located (599, 31.5%), and the surplus is the excess of identifiers over coordinates (332, 17.5%). The share is the larger because a coordinate held by three identifiers adds three to the share but only two to the surplus.

The largest co-located group is 10. We fix one convention for the whole paper, which is that **every identifier counts**, and we quantify what that choice does to the result.

Co-location is not constant across the record. It is small in the early years and grows sharply after 2016. Two measures answer to the name, and they are not the same, so we give both. The *share* at a shared coordinate is the fraction of identifiers that occupy a coordinate another identifier also occupies; over the whole record it is 31.5% (599/1899). The *surplus* is (identifiers - coordinates)/identifiers, the fraction by which the identifier count exceeds the coordinate count; over the whole record it is 17.5% (332/1899), and because it is computed within each year it rises to 22.8% by 2025, when the later co-location is concentrated. On either measure the share is the larger; the surplus is what appears in the denominator when the two units are compared, and the sense is given each time. Co-location of either kind is large enough by the later years that the choice of unit moves the headline, and the location is therefore the headline unit. The unit-and-concentration analysis reports the unit comparison year by year. The trend survives deduplication, but the magnitude is set partly by the unit: the endpoint fall is 67% on identifiers, 57% on identifiers on the fixed hull, and 45% on locations.

Local coordinates use an equirectangular projection about ($\varphi_0 = 54.97^\circ$ N, $\lambda_0 = 1.61^\circ$ W). The instruments of this network span about 66 km east–west and 98 km north–south, not the 20 km of a compact city centre, so the projection is chosen for that full extent, not for a city. At the latitude range the instruments occupy, 54.52°–55.40° N, the x-scale of the projection varies by at most 1.1% across the window, which is small relative to the effects measured; the departure from a true metric is therefore bounded and stated. The 60 km cut described above is the only spatial filter applied. It removes seven identifiers over the whole record (all far outside the UK). Those seven appear in 16 identifier–year records across the fourteen years, so the filter removes 16 of the 6570 identifier–year records and leaves 6554 identifier–year records in the analysis, before the within-year coordinate deduplication that defines the location unit. This matters for the reference area, and the estimator section addresses it directly: the bounding box of a dispersed network over a 66×98 km window is a poor density denominator, so we also fix a single geographic reference frame for every year, the *convex hull of the full record*, and report the index on that frame beside the per-year box.

Indices, estimators and uncertainty

Let N_y be the number of instruments present in year y, and let $c_{y,i}$ be the count in occupied cell i of a grid. Three grids appear in this paper and the differences matter, so we name them here and use the same names everywhere. The *bounding-box grid* is a 20×20 grid rebuilt inside each year's own bounding box; its cells are not the same pieces of ground from year to year, and it is used only in the early diagnostic figures for that reason. The *fixed grid* is anchored in geographic space over the whole record, so a cell means one place, and it is swept over 20, 30, 40 and 50 cells per side. The *rarefied series* is the fixed grid at 20 cells per side recomputed after every year is subsampled to a common number of distinct locations. All concentration results below are given on the fixed grid or the rarefied series; none is given on the bounding-box grid. We use three quantities.

Concentration. The Gini coefficient (Gini) of the occupied-cell counts (Equation (1)),

$$G_y \;=\; \frac{n_y+1}{n_y} \;-\; \frac{2}{n_y \sum_{j=1}^{n_y} c_{y,(j)}} \sum_{i=1}^{n_y} (n_y - i + 1)\, c_{y,(i)}, \qquad c_{y,(1)} \le c_{y,(2)} \le \cdots \le c_{y,(n_y)}, \tag{1}$$

where $c_{y,(i)}$ is the i-th smallest occupied-cell count and n_y is the number of occupied cells. G measures how unevenly instruments land across occupied space; it says nothing about population, welfare or need. Work

in environmental justice makes exactly this distinction, mapping exposure inequalities not instrument counts (Jiang and Ma; Alegria et al.; Karasaki et al.), and we follow it in keeping the two apart.

Clustering. The Clark–Evans index (Clark and Evans),

$$R_y \;=\; \frac{\bar{d}_y}{1/(2\sqrt{\rho_y})}, \qquad \rho_y = \frac{N_y}{A}, \tag{2}$$

where $\bar{d}_y$ is the observed mean nearest-neighbour distance and A the reference area. Equation (2) writes A without a year subscript because the headline uses the fixed full-record hull, $A = 4416\,\mathrm{km}^2$, which does not change across years; the per-year bounding box and the 95th-percentile core are reported as alternatives in the Supplemental Information, and where either is used the text says so. Under complete spatial randomness $R = 1$; $R < 1$ indicates clustering and $R > 1$ regularity, in the sense of Equation (2). Because R is proportional to the square root of density, $R = 2\bar{d}\sqrt{\rho}$, it is sensitive to how A is defined, and it scales as $A^{-1/2}$: a smaller reference area raises R.

The reference area and the core. The bounding box of a city network is a poor reference area: it is set by the single most distant instruments and therefore includes sea, open land and any outlying site, so ρ is depressed and R with it. Here the point is sharper than in a compact city, because the per-year box of this network moves: from $1056\,\mathrm{km}^2$ in 2013 to $6489\,\mathrm{km}^2$ in 2021 and back to $1880\,\mathrm{km}^2$ in 2026, so an R computed on each year's own box conflates a change in the arrangement with a change in the window. We therefore report three windows. The *per-year bounding box* is each year's own extent. The *fixed-hull window* is the convex hull of all instruments over the whole record, $A = 4416\,\mathrm{km}^2$, a single piece of ground used unchanged for every year, and it is the headline because it is the only frame on which the density denominator does not move. The *core* is a percentile rectangle, $[\text{-}x_{95}, x_{95}] \times [\text{-}y_{95}, y_{95}]$ about the projection origin, retaining roughly the central 90% of the points on each axis. The latter two are reported beside the headline in the Supplemental Information.

Edge correction. A nearest-neighbour mean computed inside a finite window is biased because instruments near the boundary have unobserved neighbours outside it. The multiscale functions are computed on the *fixed-hull* window with Ripley's *border* correction (Ripley), which is valid for a window of any shape. Writing $W_{\text{-}r}$ for the hull eroded by r — for a convex polygon, the intersection of half-planes shifted inward by r — the estimator is $K(r) = |W| \sum_{i \in W_{\text{-}r}} \#\{j \neq i : d_{ij} \leq r\} \big/ (n n_{\text{-}r})$, and $L(r) = \sqrt{K(r)/\pi} \text{-} r$. Only pairs whose reference point lies at least r inside the boundary contribute, so no pair is weighted by an unobserved neighbour. A translational weight $(1 \text{-} d_x/a)(1 \text{-} d_y/b)$ is valid only for a rectangular window and is not used here.

The single-index R values in the unit comparison are *not* edge corrected. They are reported raw because the raw ratio is the form the Clark–Evans literature names, so a reader recognises the quantity, and because the corrected and uncorrected estimators are not interchangeable: mixing them inside one table would be misleading. The bias this leaves is measured, not assumed. On synthetic patterns of known complete spatial randomness, where $R = 1$ by definition, the raw estimator returns 1.00–1.06 at the window sizes used here, and on points truncated to the 95th-percentile core it returns R within ± 0.05 of unity, so the bias is small relative to the effects measured.

Two edge-corrected reference values exist, and both are reported. *Donnelly's correction* (Donnelly) replaces the complete-spatial-randomness expectation $1/(2\sqrt{N/A})$ with an edge-corrected value derived from the window's area and perimeter. On a complete-spatial-randomness control inside this hull it returns 1.00 ± 0.07 at $N = 50$, tightening to 1.00 ± 0.01 at $N = 2000$; the raw expectation passes that same check only in the large-n limit. The reference implementation (Baddeley and Turner) is more cautious than the formula, however: spatstat refuses the correction on a non-rectangular window, on the grounds that Donnelly's coefficients were estimated by simulation on rectangles, and this hull is not one. *The general-region alternative*, spatstat's `cdf` correction, carries no rectangular assumption: it compares the observed mean nearest-neighbour distance with the mean for random points drawn in the same window.

The three readings are as follows. The raw index falls 45% (0.367 to 0.202; slope -0.0107 per year, $p = 0.037$); the `cdf` correction steepens the trajectory, to 57.2% (-0.0159 per year, $p < 10^{-4}$), though it *raises* the level rather than lowering it in the two earliest years (0.384 against 0.367 in 2013, 0.332 against 0.316 in 2014, and lower from 2015 onward); Donnelly flattens it slightly, to 41.9% (0.341 to 0.198), which is a *smaller* fall than the raw index, not a larger one. The three therefore do not agree on the magnitude, and the earlier statement in this paper that both corrections fall faster than the raw index was wrong: only `cdf` does. All three fall, and the direction is robust to the edge treatment; the magnitude

is not. We keep the raw index as the headline for comparability — it is the conventional definition and the one a reader will recognise — and not because it is better centred on randomness: on these data the `cdf` correction is, and a reader who wants the general-region value should take the 57.2% fall.

Neither correction addresses co-location, because that is a question about which records are one unit, not about the window edge. The corrections steepen the series; merging co-located registrations at 25 m is what removes the decline (the unit-and-concentration analysis). The two adjustments are independent, and the paper reports both.

Thematic imbalance. Theil's T (Theil; Barabesi et al.) over theme counts (Equation (3)),

$$T_y \;=\; \frac{1}{K_y}\sum_{k=1}^{K_y}\frac{x_{y,k}}{\bar{x}_y}\ln\!\left(\frac{x_{y,k}}{\bar{x}_y}\right), \tag{3}$$

with $x_{y,k}$ the number of instruments in theme k and $\bar{x}_y$ the count mean. Two caveats apply to T as reported here. It measures *total* imbalance over themes, not the between-theme component alone, and because the number of themes K grows across the record (from the first year's set to twelve), T rises mechanically as new themes appear even if no single theme becomes more dominant. The cross-check against an independent implementation, the known-answer controls and the analysis code are in the Supplementary Material. One convention difference is worth recording so that a reader comparing the two implementations does not mistake it for a disagreement: spatstat's `Lest` column named `border` holds $\sqrt{K/\pi}$, whereas this paper's L is $\sqrt{K/\pi}-r$. The two therefore differ by exactly r by construction, and the cross-check compares like with like by adding r back.

Uncertainty. Because the record has only fourteen annual points, every procedure is chosen for its behaviour at that length and each claim is carried by more than one of them. *Resampling* (Efron) over the instruments, 1000 draws, gives the per-year concentration intervals and the rarefied series. *Slope inference* uses ordinary least squares, a permutation test that shuffles the annual values across years and is exact under exchangeability, Mann–Kendall (**(year?)**) and Theil–Sen (**(year?)**), together with the HAC family — Newey–West (**(year?)**), a circular moving-block bootstrap, the fixed-b approximation of Kiefer and Vogelsang (**(year?)**) and a Prais–Winsten fit (Prais and Winsten). The HAC family is reported as a sensitivity and not as primary evidence, because at fourteen points the OLS-plus-Newey–West test with a $t(12)$ critical value is anti-conservative, rejecting 16.3 % of the time at a nominal 5 % even on pure noise and 19.3 % at the autocorrelation the headline series shows; the Supplemental Information gives every series under every procedure and simulates each test's size. *Equivalence*, not non-rejection, distinguishes a null that is evidence of absence from one that is merely an absence of evidence, so every null carries a two-one-sided-tests procedure (Schuirmann; Lakens et al.) against a reference margin taken from the effect the uncorrected index shows, with half of that margin reported as the stricter reading, and a minimum detectable effect at 80 % power in place of a post-hoc power computed at the observed slope. *Cluster inference* is against an *inhomogeneous* null that preserves the first-order intensity, because complete spatial randomness mainly restates that a city network is not uniform, which is not in question. *Rarefaction* subsamples every year to a common sample size before a trend is read from it, with both the Newey–West and the rank-based Theil–Sen interval reported; it is applied to the count-based indices and to the clustering series, and the unrarefied trends are reported beside them in Table 2 rather than replaced by them.

Footprint matching. One further procedure is used and is defined here because several results depend on it. A fixed window and a per-year window are both unsatisfactory: the first holds the area constant but lets the support mismatch between early and late networks stand, and the second makes the area a moving function of the network being measured. *Footprint matching* removes both by asking what the present network would look like if it occupied an earlier year's ground. For each year y we take that year's own convex hull, select the present network's locations that fall inside it, and compare their multiscale function with that of year y's own points at the same sample density. The window and the sample size are therefore held to the year being judged, and only the arrangement can differ. Eleven of the fourteen years yield a footprint containing enough present-day locations to test; the three that do not are the largest late-year hulls, which contain almost the whole network and so reproduce the fixed-hull comparison. The per-year series is tabulated in the Supplemental Information.

Search and analysis choices. Fitting a trend to a record of this shape offers many defensible forks, and the ones taken here are stated, not presented as choice-free: the reference area (three windows), the unit (identifiers and deduplicated locations), the estimator (ordinary least squares, HAC, moving-block,

fixed-b, Prais–Winsten, Theil–Sen), the null in the infill test, and the rarefaction anchor (39, 150 and 200 locations). The *sign* of the clustering change is negative under every reference frame and unit, which is the fork that would have mattered most, but its magnitude runs from 45% on the primary unit and frame to 67% on the largest, so the magnitude is not a property of the network. The full set of procedures is tabulated in the Supplemental Information.

Multiplicity is controlled with the Benjamini–Hochberg procedure (Benjamini and Hochberg) across the per-year clustering tests. These are thirteen consecutive year pairs, not fourteen years, because each pair is tested against the pooled positions of that pair alone; the correction is therefore applied to a family of thirteen and is approximate, since consecutive tests share a year. Ten of the thirteen survive at a 5 % false-discovery rate: 2014, 2015, 2016, 2017, 2019 and the five years from 2022. The three that do not are 2018, 2020 and 2021, and the same three are the years whose new positions sit inside the pooled-subset interval at the single-test level, so the correction removes exactly the years the uncorrected test already failed to separate; every year pair's raw and adjusted p-value is tabulated in the Supplemental Information under "Multiplicity, the random seed and the code". Across the paper as a whole about 182 fitted comparisons appear, counting the level series by frame, unit and year, the Gini by resolution and year, the rarefaction anchors, the interval procedures and the bandwidth sweep. This larger family is not corrected as a single test — many of its members are descriptive readings of one series under different frames not independent hypotheses — but the count is stated, so a reader can see the size of the search. All random draws use one fixed seed (`20261003`). The seed enters the rarefied series through the subsampling draws, so it moves those fitted slopes; the check below bounds that. Over three seeds the rarefied R slope on the fixed hull is $+0.0003$ (se 0.0023), $+0.0003$ (0.0022) and $+0.0005$ (0.0022), a spread of 0.0002 per year, with the interval containing zero under every seed and the endpoint values moving by at most 0.002. The seed therefore affects the fitted rarefied slopes by less than 0.0003 per year, and no reported conclusion depends on it; the text quotes the stated-seed value where a slope differs in the third decimal.

Isotropy: the one symmetry test the record can support

The framing above defines every quantity here as a departure from a reference state that is symmetric in a stated sense, so it is worth asking whether the record departs from a symmetry that can actually be tested. Rotation is the one available. A homogeneous Poisson process is invariant under rotation about any point, and a deployment biased towards particular street orientations is not, so a directional version of K separates them directly. We compute the orientation of every pair within 1 km of one another for the 2026 network, undirected so that 0 and π are the same direction, and bin the orientations into 8 sectors of $\pi/8$. The unit is deduplicated locations, as everywhere else, so an instrument registered twice contributes one point.

The pairs are not uniformly distributed over direction: a chi-square against a uniform distribution over the eight sectors gives $\chi^2 = 81.3$ on seven degrees of freedom ($p < 10^{-14}$), and the ratio between the best- and worst-populated sector is 1.42. But that chi-square treats 5590 pairs as independent observations, and they are not—every point appears in many pairs—so it is a description of the pattern and not a calibrated test, and we do not read a p-value from it.

The calibrated test is a *rotation null*. An irregular window constrains which pairs can exist at all, so the appropriate question is not whether the orientations are uniform but whether they differ from what the observed pattern would give under a different orientation inside the same window. We therefore rotate the 2026 pattern about its centroid, keeping the window, the clustering, the pair count and the number of points exactly as observed, and recompute the sector counts over 999 rotations. Under that null the best-to-worst ratio has mean 1.37 (sd 0.06), and the observed 1.42 falls inside its 95 % envelope, with 1 of the eight sectors outside. The network's orientation, relative to the window it is measured in, is therefore indistinguishable from a random rotation of itself at this radius.

That null also settles a claim the geometry alone might suggest. It is tempting to attribute the sector imbalance to the elongated hull, but the window is nearly orientation-blind at this scale: a dense complete-spatial-randomness sample of 60 000 points inside the same hull reproduces a best-to-worst ratio of 1.01, against the observed 1.42. The window therefore does *not* account for the anisotropy, and the earlier view that it did was wrong. What we can say is narrower and better supported: the pattern is non-uniform over direction, and it is not non-uniform in a way that differs from a rotated copy of itself, which is a statement about the pattern's orientation relative to the frame rather than about isotropy in general.

Interpretive boundaries

Following the practice of declaring what a departure does *not* license, we set out the limits before presenting results.

Table 1. Reference state and interpretive boundary for each measured quantity.

Component	Symmetry reference state	Boundary of interpretation
Concentration (G)	Equal instrument counts across occupied cells	Does *not* represent population served, need, welfare, environmental burden, or any normative judgment about where instruments *should* be.
Clustering (R)	Complete spatial randomness, $R = 1$	Does *not* imply a coordination mechanism, a policy cause, or that clustering is undesirable. It *is* sensitive to the study-area definition, which is why the 95th-percentile core is reported separately.
Thematic imbalance (T)	Equal counts across themes	Does *not* imply that a theme is over- or under-measured relative to need; a city may require more traffic counters than seismometers.
Presence in year y	Instrument appears in one or more files that year	A silent instrument is *indistinguishable from* a decommissioned one, and presence does not imply continuous operation within the year.

RESULTS

What the paper measures, and what it finds

Table 2 collects the headline quantities on the frame and unit each is stated on, because both move the magnitude: the frame sets the density denominator, and the unit decides whether a re-registered device counts twice. Its p-values are nominal OLS values. At fourteen points the same simulation puts the OLS test at 4.7 % rejection under independent errors and 8.5 % at the lag-1 autocorrelation the fixed-hull series has (+0.15), while the HAC test is anti-conservative throughout. The section then does one thing: it separates the apparent change in the Clark–Evans index into the parts that are measurement and the part, if any, that is geography.

Table 2. Headline results, with the frame and the unit each is stated on. The p-values are nominal OLS values.

Quantity	Frame	Unit	First	Last	Change	Test
Population	—	identifiers	41	706	17.2×	census
Population	—	locations	39	561	14.4×	census
Clustering R	fixed hull	locations	0.367	0.202	-45%	OLS $p = 0.037$
Clustering R	fixed hull	identifiers	0.357	0.155	-57%	OLS $p = 0.018$
Clustering R	per-year box	identifiers	0.729	0.238	-67%	(moving frame)
Clustering R	merged 25 m	sites	0.367	0.351	-4.3%	OLS $p = 0.74$
$\ln \bar{d}$ vs $\ln N$	fixed hull	locations	exponent -0.603		vs $-\frac{1}{2}$	$t =$ -1.52

The census: a population that turns over, not one that accumulates

The population rose from 41 instruments in 2013 to 706 in 2026 (17.2×; Fig. 2), but not as a ramp. Of the 1906 identifiers that ever appear, 1200 make their last appearance before 2026, and **562 leave and return**: they are absent for at least one year between their first and last appearance, 389 appear in a single year only. Roster entries reach 2589 against 1899 distinct identifiers, and the excess is exactly these returning instruments, so no departure count in this paper can be read as a decommissioning rate. Two definitions are in circulation in an archive of this kind, and we name whichever we use: *first and last appearance* are lifetime events, and *entry to and exit from the annual roster* are membership events, which differ because an identifier can leave and return. We report a *replacement rate* for the latter: the share of arrivals and departures that swap places instead of adding to the network. Its median over the record is 0.58, lowest in the growth years (2015, 0.20) and highest in the flat ones (2026, 0.99, where 60 arrivals and 61 departures

nearly cancel). Table 3 gives the series, with a gap-filled panel as the sensitivity check; filling every one-year gap removes most of the apparent churn in 2018 (672 against 367 as published). We do not report the growth benchmark $|\Delta N|/2N$ as a finding, because the excess of turnover over it is algebraically $\min(\text{in}, \text{out})/N$, a positive quantity in any year with both an arrival and a departure, and therefore an identity and not evidence. First and last appearances bound instrument lifetime observationally and do not measure it; we use them only to establish that the population is dynamic and that its growth path changes composition.

Table 3. Annual membership churn as a replacement rate, with a second panel that fills one-year reporting gaps. Both panels begin at 2014, since 2013 is the founding year.

All 1899 *identifiers, as published*				
Year	*N*	In	Out	Replacement
2014	97	69	13	0.32
2015	228	147	16	0.20
2016	199	124	153	0.90
2017	605	500	94	0.32
2018	367	163	401	0.58
2019	474	327	220	0.80
2020	749	350	75	0.35
2021	800	151	100	0.80
2022	566	122	356	0.51
2023	466	102	202	0.67
2024	549	173	90	0.68
2025	707	260	102	0.56
2026	706	60	61	0.99
One-year gaps filled				
Year	*N*	Turnover	Replacement	
2018	672	0.188	0.83	
2021	846	0.113	0.62	
2026	706	0.074	0.84	

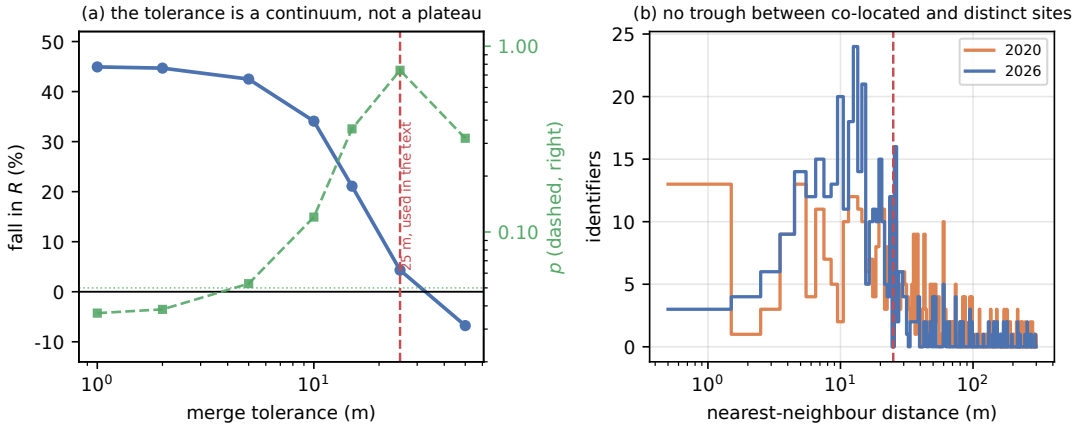


Figure 1. The merge tolerance. (a) Fixed-hull R and its OLS p-value against the merge radius; the dashed line is the 25 m row of the tables. (b) Nearest-neighbour distances for 2020 and 2026.

The apparent clustering trend is a measurement effect

This is the paper's central result and it is negative. On the fixed-hull window, in deduplicated locations, the Clark–Evans index falls from 0.367 in 2013 to 0.202 in 2026, a 45% reduction. The direction is the same under every frame and unit, so it does not depend on that choice; the magnitude differs by more than twenty points across them. We show that the fall is produced by three things that are not geography — densification, co-located registration, and the reference window — and that once all three are accounted

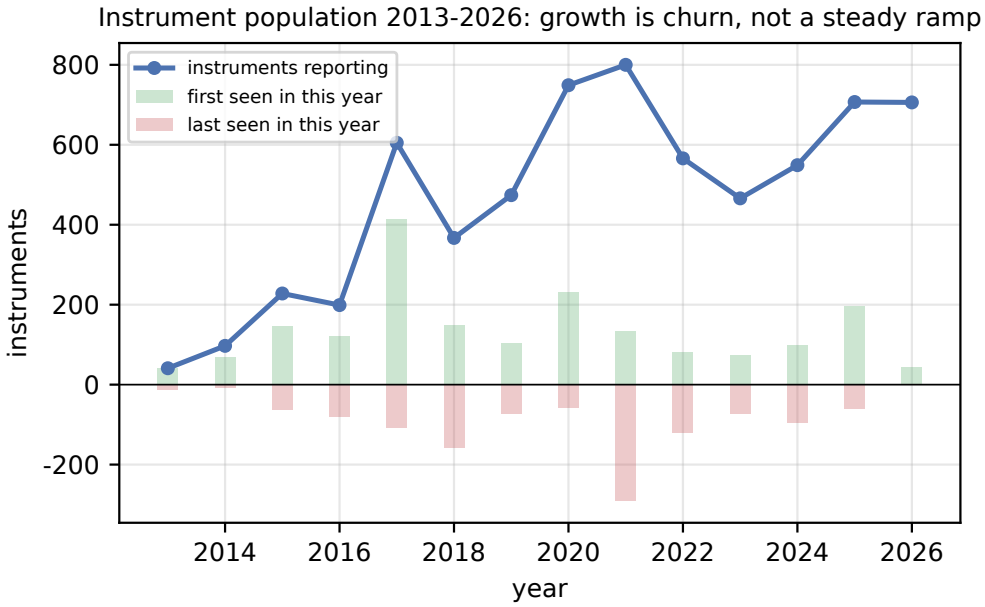


Figure 2. Instrument population, 2013–2026, with each instrument's first and last appearance.

for no trend in the nearest-neighbour geometry is detectable. Figure 3 gives the series on the fixed hull under both units.

The window. R scales as $A^{-1/2}$, so a smaller reference area *raises* R. The per-year bounding box moves by a factor of six across the record, so the box series is moved by the window in both directions: the 2013 box is 1056 km^2, a quarter of the hull, and the square root of the area ratio ($A_{\text{hull}}/A_{\text{box}} = 4.18$) is 2.04, which is the ratio of the two 2013 values (box 0.729, hull 0.357). By 2021 the box has grown past the hull (6489 against 4416 km^2) and the box value is *below* the hull value. Quote either and the level is a property of the frame. We fix the hull, which is the only frame on which the density denominator does not move, and report the box beside it.

The window has a second form: the support mismatch. The fixed hull is the union of every year's points, and the early years occupy a small part of it — the 2013 footprint is 671 km^2, 0.15 of the hull, and the 2021 footprint is 1.00 of it (Fig. 4b). Measured as bounding boxes, the 2013 network occupies 23.9 % of the reference frame and the 2026 network 42.6 %. A pattern confined to a fraction of its window has its K inflated by roughly the inverse of that fraction, so the early years carry a support penalty that the later years do not. Recomputing every year on *its own* hull, which matches the density denominator to the pattern's extent, removes most of the decline: the 1 km multiscale series becomes $5505, 2598, 2291, 4364, \ldots, 2114$ in place of $13640, 7673, 5251, 8991, \ldots, 4028$, and its fitted trend falls from -426 m per year (se 110) to -99 m per year (se 107), which is not distinguishable from zero. The same defect therefore inflates the box series in one direction and the fixed-hull series in the other, and neither is free of it.

The unit. The exact-coordinate deduplication used above collapses only identifiers at the *same* coordinate; two registrations a few metres apart remain two "locations". Co-location is a property of the publishing schema, not of the deployment, and it is large in the late years: 74.2% of 2026's identifiers have a neighbour within 25 m and 39% within 5 m, against 9.8% in 2013. Merging identifiers at that radius takes the fall to 4.3% (slope +0.0014, $p = 0.74$). The result is monotone in the merge radius with no plateau, so the radius is a choice, not a threshold the data identify: 44.9% at 1 m, 42.5% at 5 m, 34.1% at 10 m, 21.1% at 15 m, 4.3% at 25 m, and -6.8% — a sign reversal — at 50 m (Fig. 1a; Table 6). What would justify a threshold is a trough in the nearest-neighbour distribution separating co-located registrations from distinct installations; there is none, the largest gap below 200 m being 35 m in 2014 and 6 m to 26 m in every other year, while the pooled distribution is smooth across four orders of magnitude (Fig. 1b). The merge is not an artefact of the linkage rule either: complete-linkage gives 4.7% ($p = 0.774$), a centroid representative gives 4.3%, and the largest cluster single-linkage forms has a diameter of 74 m against a median of 12 m, so chaining is bounded. Grid-snapping instead gives 31.1% ($p = 0.146$), reported as a

limitation, not a footnote. Since the radius is not identified by the data, none of these values has priority: they bound the effect, and do not estimate it.

The controls interact, so they do not decompose into shares. It would be natural to read the paragraphs above as three independent corrections and ask what fraction of the 45% decline each accounts for. That reading is wrong, and the arithmetic says so. On the fixed hull, merging at 25 m removes the decline (4.3%). But the support correction goes the *other* way: measuring each year on its own hull *enlarges* the decline to 64.7%, because a small early footprint raises the early-year index. Combine them and the decline returns: on the merged unit and each year's own hull the fall is 38.6% (slope -0.0201, $p = 0.004$). The four configurations are set out in Table 4, and they do not add up in any direction—one control removes the effect and the other restores it. Registration and support are therefore reported as a *sensitivity range* (45% down to 4.3% on the fixed hull, and 64.7% under support matching), not as a partition. Equation (4) likewise decomposes the *index* into a distance term, a count term and a window term, and cannot express the registration factor at all, since registration is not among its terms; it is an identity, useful because it shows which term is on the right-hand side of the regression, not a bookkeeping of the 45%.

Table 4. The two controls in combination, on the location unit. They do not add up: registration alone removes the decline, and support matching restores it.

Unit	Support window	Slope/yr	SE	OLS p	Fall
Identifiers (published)	fixed hull	-0.0126	0.0046	0.018	56.4%
Identifiers	**fixed hull**	-0.013	0.0046	0.018	57%
Locations	fixed hull	-0.011	0.0046	0.037	45%
Merged 25 m	fixed hull	+0.0014	0.0041	0.742	4.3%
Merged 25 m	own hull	-0.0201	0.0057	0.004	38.6%
Exact locations	own hull	-0.0382	0.0059	< 0.001	64.7%

The robust estimator sharpens the diagnosis and does not rescue the result. Replacing the mean nearest-neighbour distance with the median, against its own complete-spatial-randomness expectation, gives a fall of 94%, against 45%. That is the opposite of a rescue. A robust statistic that moves *more* than the mean is telling us the change lives in the body of the nearest-neighbour distribution, and for the late years that body is the co-located mass. Under the merge the median behaves as the mean does (44.5% at 25 m, $p = 0.32$), so it is not a substitute for the merged unit. We report it as a diagnostic of where the change sits, not as a headline estimator.

Rarefaction. Subsample every year to a common sample size and the trend disappears: the rarefied location slope is +0.002 per year with a Newey–West interval of $[-0.002, +0.006]$ containing zero, against -0.011 unrarefied. The null is informative and not merely underpowered — the minimum detectable effect at 80 % power is 0.008 per year, well below the unrarefied slope — and an equivalence test rejects at $p = 0.001$ within the unrarefied magnitude. Its limit must be stated: with 39 locations the nearest-neighbour distance is set by separation at kilometres, so the null rules out a coarse-scale trend and says nothing about the fine scale at which the co-location operates. Table 5 gives the series year by year.

Table 5. Clark–Evans R on the fixed full-record hull, unrarefied and rarefied to a common sample size ($N = 41$ identifiers, $N = 39$ locations).

		Rarefied R		Unrarefied R	
Year	N (ids/locs)	Identifiers	Locations	Identifiers	Locations
2013	41 / 39	0.357	0.367	0.357	0.367
2017	605 / 518	0.369	0.391	0.233	0.258
2021	800 / 729	0.461	0.466	0.407	0.429
2026	706 / 561	0.352	0.382	0.155	0.202

The index identity, which is the cleanest quantitative result here. On a fixed frame

$$\ln R \;=\; \ln \bar{d} \;+\; \tfrac{1}{2}\ln N \;-\; \tfrac{1}{2}\ln A \;+\; \ln 2, \tag{4}$$

so the apparent trend in R is the sum of a distance term, a count term and a window term. This is an algebraic identity, not an allocation: it says which term of the index a regression moves, and it has no

registration term. We verified the identity to 10^{-15} and fit the distance term directly. On the location unit $\ln \bar{d} = \text{const} + \hat{\beta} \ln N$ with $\hat{\beta}$ = -0.603 (se 0.068), against the $\beta = -\frac{1}{2}$ that densification at constant shape forces: the difference is 0.103, t = -1.52, $p = 0.15$, so the densest years are somewhat more clustered than density alone predicts but not significantly so. On the identifier unit the same fit gives -0.651 (se 0.086), $|t|$ = -1.75, which is the value the published benchmark reports without naming the unit. On the merged unit the exponent moves towards the structural value (-0.550 at 10 m, -0.426 at 25 m), the same registration effect appearing a third time. Random thinning of the present pattern gives -0.745 (se 0.066) over twenty draws, so the observed exponent lies between the densification value and the thinning value and is not distinguishable from either. This exponent is the paper's one clean measurement and is not attributed to a cause.

Two results here are established, not asserted. *Registration*, not geography, produces the co-location, and the archive itself says so: the collapse is confined to one of the two published coordinate resolutions and is unchanged when the other is rounded to the same quantum, so the co-located records are duplicated registrations, not positions quantised into coincidence (the co-location analysis). *The apparent clustering trend*, once that and the window are controlled, leaves no trend in the nearest-neighbour geometry (the clustering-trend analysis). The first names a cause for the measurement effect; the second removes it from the geography. We do not extend that cause to the deployment.

The unit of analysis, and the concentration index

Co-located identifiers are an artefact of the publishing schema, not of geography, and removing them changes the headline, so the location is the headline unit and the identifier figure is reported beside it. Of the 1899 identifiers that enter the spatial analysis, 1567 occupy distinct coordinates; 599 of them, or 31.5% of the population, sit at a coordinate another identifier also occupies, and the surplus is 332 or 17.5%. The identifier series is not wrong to report — it is what the platform counts — but it double-counts the schema, so it is the secondary figure. *Concentration is a sample-size effect in the same way.* The Gini of occupied cells rises at 0.013 per year uncorrected, and rarefying every year to a common sample size reverses the sign (-0.004 at 20 cells per side; -0.0068, -0.0106 and -0.0076 at 30, 40 and 50). The two answers are of opposite sign and neither is negligible, so the index does not identify a direction of change, and at fine resolutions it is near-degenerate — at 50 cells per side the smallest sample occupies 24 of 2500 cells and most counts equal one. No claim about concentration is made in either direction. The grid resolution, grid origin, gap filled roster and core-definition sweeps are in the Supplemental Information.

Table 6. The fixed-hull location series under tolerance deduplication: single-linkage at radius r, one identifier kept per cluster.

Deduplication	2013 R	2026 R	Fall	Slope/yr	OLS p	2026 z
Exact coordinate (published)	0.367	0.201	45.3%	-0.0108	0.035	-6.6
Within 5 m	0.367	0.211	42.5%	-0.0101	0.053	-6.8
Within 10 m	0.367	0.242	34.1%	-0.0076	0.120	-6.0
Within 25 m	0.367	0.351	4.3%	+0.0014	0.742	-3.6

The co-location is duplicated registration, not a rounding artefact

The merge radius would be a free choice only if the co-location had no origin we can identify. It has one, and the archive records it. Coordinates are published to two different resolutions: 36.8% of identifier–year records are written to six decimal places or fewer, a quantum of about 0.11 m in latitude, and 42.8% to twelve places or more, where the printed value is sub-millimetre. These are two publishing paths, not two kinds of instrument.

Pooled over the record, 62.5% of high-precision records have a neighbour within 25 m against 32.6% of low-precision records, so co-location is roughly twice as common where coordinates are published at full resolution. The comparison does not hold in that direction for every year. In the later years it inverts: on the sub-metre scale, 73.2% of low-precision records in 2026 have a neighbour within 0.5 m against 8.3% of high-precision records, the reverse of the pooled reading. The two classes therefore do not differ by a constant factor; whichever path co-locates changes over the record, and the merge tolerance crosses that change instead of sitting on one side of it.

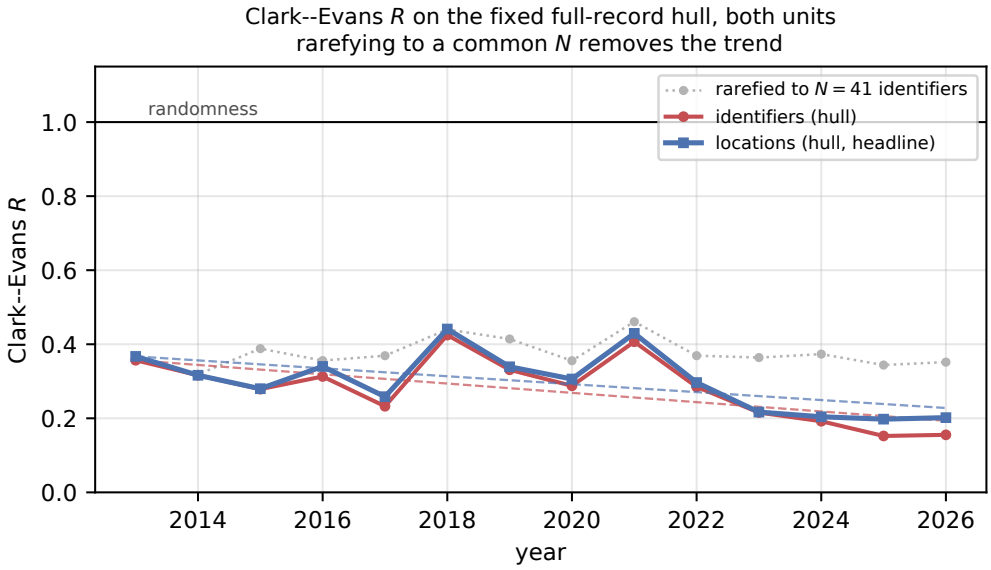


Figure 3. Clark–Evans R by year on the fixed full-record hull, both units, with the rarefied series and the fitted trends. The band is the homogeneous CSR envelope.

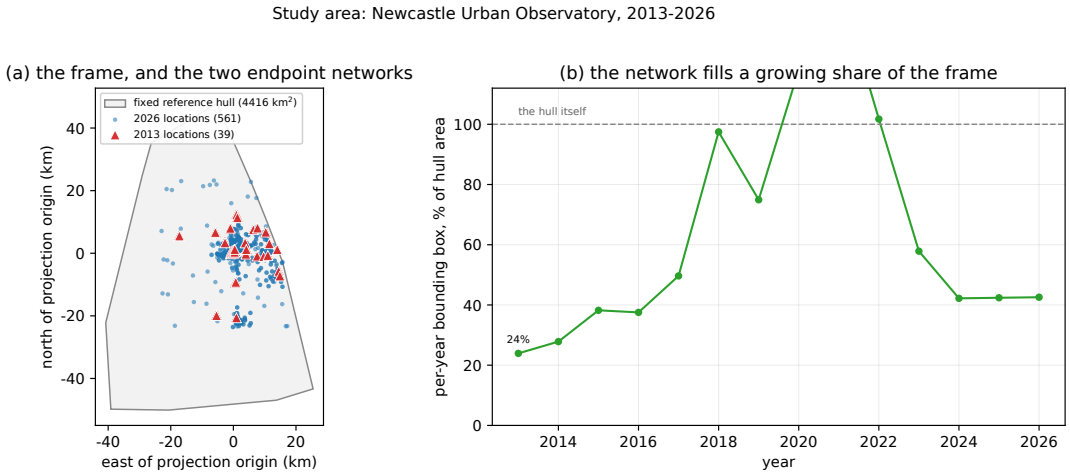


Figure 4. The study area. (a) The fixed reference hull, with the 2013 and 2026 networks on it. (b) Each year's bounding box as a share of that hull.

What survives all of this is a narrower statement, and it is the one that identifies the origin. Rounding the high-precision class to the low-precision quantum leaves its co-location unchanged: 2017 stays at 17.3% and 2020 at 14.8%, while the same years' low-precision class sits at 3.8% and 0.9% (Table 7). The co-located pairs are therefore already coincident to better than a tenth of a metre, in the class where rounding could not have created them. The co-location is duplicated registration in the archive, not precision lost in publication.

It also survives inside one class, but only weakly, and the qualification matters. Restricting to high-precision records only and holding the sample size fixed at 33 by random subsampling over the eleven years for which that class is populated, the fall is 34.3%. The matched control — random subsets of the pooled high-precision locations at the same size — has a mean fall of 8.1% with a 95 % interval of $[-56.0, 56.8]$ over 200 draws. The observed 34.3% therefore lies *inside* that interval: it sits at the 84.5th percentile of the control distribution, which is suggestive but not significant at the 5 % level. We report it as consistent with one path carrying part of the effect without the mixing of two paths, not as an independent confirmation of it, because the interval that would have to be excluded spans both signs.

The collapse is not uniform across years, and it follows registration, not geometry. In 2025 and 2026 the low-precision class collapses to 43.6% and 44.7% while the high-precision class sits at 9.1% and 3.6%; in 2017 and 2021 the ordering reverses, the high-precision class at 17.3% and 14.9% against 3.8% and 0.3%. The identity of the co-locating path therefore changes across the record, and the merge tolerance crosses that change. This is why the curve has no plateau. The radius is not locating a physical separation between co-located and distinct installations; it is progressively removing whichever path co-locates first, and the path that does so is not the same one throughout. The 25 m value is the smallest radius at which neither path shows a substantial collapse, and it is a bound on the artefact, not a measurement of the deployment.

Table 7. Exact-coordinate collapse by published coordinate precision, 2017–2026. Rounding the high-precision records to the low-precision quantum (0.11 m) leaves the value unchanged, so the co-location is duplicated registration.

	High precision (12 or more dp)		Low precision (6 or fewer dp)	
Year	*N*	Collapse	*N*	Collapse
2017	283	17.3%	182	3.8%
2020	317	14.8%	320	0.9%
2021	343	14.9%	374	0.3%
2025	350	9.1%	257	43.6%
2026	336	3.6%	284	44.7%

What survives: a multiscale level, not a trend

Against an *inhomogeneous* null that preserves each year's first-order intensity — a kernel-smoothed surface fitted to that year's own points and resampled 199 times, with the 750 m bandwidth stated because the result depends on it — the multiscale function $L(r)$ exceeds the entire envelope at 250 m to 1.5 km in all fourteen years, at 100 m in thirteen (the exception being 2013, whose 39 locations cannot resolve that scale) and at 2 km in twelve, and it never falls below it at any scale in any year. That is a stronger statement than the single-index version of the same test, because L uses the border-corrected estimator and the location unit removes the co-located channels. The surface is fitted to the points it tests, so the separation is a description, and the scale range over which it is unanimous is stated.

Its *decline*, however, does not survive support matching, as the clustering-trend analysis shows. Note first that this fall is not the same movement as the one in the nearest-neighbour index, and the two are worth naming separately because they run in opposite directions at their own scales. A *falling R* means nearest neighbours sit closer than the density predicts, that is *more* short-range clustering; a *falling L* at 1 km means fewer excess neighbours at that scale, that is *less* medium-scale clustering. The two indices are therefore not two readings of one change, and this paper's subject is the apparent strengthening of the *nearest-neighbour* geometry, which is the index that does not survive. Wherever this paper says "the decline" without naming an index, R is meant.

What survives for L is not a trend but a level: on footprint-matched windows the 2013 network is more clustered at 1 km (5505 m) than the present network is inside the same footprint at the same density

(envelope 502–1986 m), and nine of the eleven testable years lie above their own envelope. The early network differs from the present one at medium scales; the record does not show a continuing drift, and the fitted slope halves when 2013 is removed (-269 m per year, $p = 0.019$, against -426 over all years). The footprint-matched procedure is defined in the estimator section and tabulated in the Supplemental Information: for each year, the present network's locations that fall inside that year's own convex hull are compared with the multiscale function of the year's own points at the same density, so the window and the sample size are held to the year being judged.

Size and time cannot be separated, and the infill question is open

The association between clustering and size looks strong on the per-year window ($\ln R = 1.163 - 0.369 \ln N$, $R^2 = 0.849$) and does not survive fixing the window: at a fixed area the size term is dominated by the identity of Equation (4), the exponent is not distinguishable from the structural value, and the rarefied series has no time trend either — subject to the scale limit stated where rarefaction is introduced, that at 39 locations it resolves only structure above about 5 km, so it excludes a coarse-scale time trend and not a fine-scale one. Neither size nor elapsed time is identified, so the change is attributed to neither.

One diagnostic that might have attributed it does not. If new instruments filled the gaps *between* existing ones, the network would be densifying inward; if they arrived outside, it would be expanding. We tested this by pooling each year pair's locations and permuting the new/existing labels, which holds the spatial support and the intensity fixed. On the published unit new locations lie *further* from the existing set than the permutation holds, in 12 of 13 year pairs, and they also cluster among themselves relative to a pooled-subset null in 10 of 13 pairs — consistent both with outward expansion and with batches deployed into contiguous new ground, which this record cannot separate. Both counts halve once co-located registrations are merged (7 of 13 and 6 of 13), so even this diagnostic is a statement about the published unit. We therefore report no mechanism.

Two publication discontinuities

The platform changed what it published twice, and both changes are visible in the archive rather than in the city. Vehicle share collapses to 16.8% in 2018 on identifiers (17.4% on locations) because the vehicle files for 2017 carry `Cars` and `Parking Spaces` and the 2018 files do not; a reader using 2018 as representative of that theme would be reading the publishing schema. Archive file counts run 11–75 per year to 2025 and then jump to **254** in 2026, when the platform began publishing monthly files, not bundled ones, while the instrument roster is continuous (706 against 707), so the discontinuity is in file granularity and not in the deployment. Neither propagates into the fitted trend: on the fixed hull the slope moves from -0.011 to -0.010 across the four exclusion samples, a range of 0.0011 against an interval of 0.016 for those coefficients, and on the box frame it is unchanged to within 8%. The final year covers nine months, not twelve, and every headline result is reported with and without it: dropping it leaves the permutation test marginal ($p = 0.101$) while Mann–Kendall, which depends on rank order, not year values, still rejects ($p = 0.030$), so the decline is better described as monotone than as linear. The theme composition, the gap-filled roster, the core sweep and the exclusion sweeps are in the Supplemental Information.

DISCUSSION

What this means for anyone using this network

A study that uses this observatory at a single moment is using one geometry from a series whose size grew 17.2× and whose apparent clustering fell by between 4.3 and 67 per cent depending on how co-located registrations are counted and which window is used. The practical warning is therefore not a number but the dependence itself: a spatial result computed against this platform is conditioned on the year, the frame and the unit, and that conditioning is not usually reported. The same caution applies to derived products built on open platform data, including the interoperability layers and digital-twin interfaces that aggregate a platform without re-censusing it (Anthony), and to the frozen snapshots used as execution targets; one such snapshot is audited here, and the audit is in the Supplementary Material. This is the same family of problem **(author?)** (Xia et al.) identify for carbon inventories: an apparent structural property that depends on how the measurement was constructed. Their construction variable is accounting choice; ours is the definition of the unit and the frame.

A transferable rule for longitudinal spatial reporting

The specific numbers above belong to one city. The argument that produced them does not, and this is the part of the paper we expect to outlive the case. Wherever a sensor platform re-registers instruments under new identifiers, and wherever a longitudinal study reads a density-dependent spatial index, the same three quantities condition the answer, and each can be reported at no cost to the analysis:

1. **The support of each year's pattern, beside the window it is measured in.** A window inherited from the union of the record penalises its early years by a factor near the inverse of the area they occupy — here 0.15 of the hull in 2013 — and inflates them in the other direction as the window is fixed. Reporting the occupied area per year makes the penalty visible; measuring on each year's own footprint removes it.

2. **The degree of registration co-location, as a distribution, not a flag.** The share of identifiers with a neighbour inside a stated radius suffices, reported per year. One number states the problem and its growth; a binary "deduplicated" flag hides both, and cannot distinguish an archive that is clean from one where co-location arrived with a schema change. Where the archive publishes coordinates at more than one resolution, the share should be reported *per resolution*, because the classes need not co-locate alike and here they do not: one path collapses to 44.7% while the other sits at 3.6% in the same year. That single split is what identifies the artefact as duplication, not geometry, and it is available before any clustering is computed.

3. **The rarefied and unrarefied series together.** A count-based index confounds a change in arrangement with a change in population, and the two answers can carry opposite signs, as they do for the concentration index here. Reporting both separates a genuine rearrangement from a sample-size effect.

None of the three requires a new estimator, a model, or a judgement about the deployment, and together they convert an unreportable conditioning into a stated one. The broader point is that for a growing sensor network the measurement protocol is not fixed while the object changes; the window, the sample size and the identity of the unit all move with the network, so a longitudinal statistic is stable only if those three are held and reported. That claim is independent of this city, and it is the reason the factorial comparison in the clustering-trend analysis is worth performing, not the trend worth quoting.

The headline is a floor on the measurement effect, not a ceiling. The 45% decline is measured on a window fixed across years. Measuring each year on its own hull instead *enlarges* it to 64.7% (0.942 to 0.332), because a small early footprint raises the early-year index: R scales as $A^{-1/2}$, so the 2013 network's 671 km^2 footprint lifts its index well above its fixed-hull value. Support matching and the per-year box therefore both move in the same direction, and the fixed hull is the conservative choice. An earlier version of this paper described the fixed-hull figure as an upper bound set jointly by the three factors; that description was wrong in direction and is corrected here. What the three factors do is reduce the decline when they are controlled one at a time (registration to 4.3%, the support-matched multiscale slope to zero), not bound it from above.

Limitations

The record has fourteen annual points with size and year collinear, so nothing here is an attribution. Five further limits bear on specific claims. *The merge tolerance is not identified by the data* (the clustering-trend analysis), so the effect is bounded, not estimated, and grid-snapping gives a different answer again (31.1%). *The rarefied null* resolves only structure above about 5 km, so it rules out a coarse-scale time trend and says nothing about the fine scale at which the co-location operates. *The window is not neutral*: the fixed hull is the union of every year's points, so a few outlying sites set the denominator for all years, and the earliest years carry a support penalty from which no density-dependent index is free. *The HAC intervals are anti-conservative at this length*: simulated on pure noise at fourteen points the OLS-plus-Newey–West test rejects at a nominal 5 % about 16.3% of the time, rising to 19.3% at the lag-1 residual autocorrelation the headline series actually shows (+0.15) and 29.4% at $\rho = 0.5$. The plain OLS test is a different case and the distinction matters: it is correctly sized in the same simulation, rejecting 4.7 % under independent errors, and mildly optimistic at the series' own autocorrelation (8.5 %). So the fixed-hull p-values are neither safely conservative nor badly inflated, and we carry each claim on more than one procedure rather than treating any single p-value as calibrated. *The concentration index*

of occupied cells is not usable at fine resolutions: rarefied, it is negative at every resolution with a rank interval excluding zero while the uncorrected series is positive, so rarefaction reverses its sign, and at 50 cells per side the smallest sample occupies 24 of 2500 cells, so most counts are one. Concentration is unresolved in either direction. The thematic composition, the gap-filled roster and the core-definition sweep are sensitivities in the Supplemental Information, and none of them changes a conclusion above.

CONCLUSIONS

We set out to ask how much of the apparent strengthening of clustering in the Newcastle Urban Observatory is a property of the deployment, not of the measurement. Conditional on the merge tolerance the answer is: almost none of it. On the exact-coordinate unit that the paper leads with, and on a window fixed across years, the Clark–Evans index falls 45%. Merging registrations within 25 m — a radius the data do not identify, so that figure is a bound and not an estimate — leaves 4.3%, monotonically in the radius and with no plateau. The honest reading is therefore a range rather than a point, and a reader who takes co-located registrations to be one instrument should read the decline as small.

No part of the 45% is geography that has been shown to move. The reference window moves the level in both directions and has a second form as a support mismatch between the early footprints and the window they are measured in, which *enlarges* the decline to 64.7% rather than reducing it; the count term is on the right-hand side of the index by construction; and co-located registration, which is a property of how the platform records instruments, removes the decline outright. The three do not partition the 45% into shares, and we do not present them that way: they interact, and registration and support matching pull in opposite directions (Table 4). What the combination establishes is that there is no version of the measurement, on any unit or window we tested, in which most of the 45% is left as geography once registration is controlled. Once all three are controlled, no trend in the nearest-neighbour geometry is detectable: rarefaction to a common sample size removes it, and the mean nearest-neighbour distance scales with population at -0.603 (se 0.068), not distinguishable from the $-\frac{1}{2}$ that densification at constant shape forces. The multiscale decline is largely a support artefact as well, falling from -426 to -99 m per year when each year is measured on its own footprint.

What survives is smaller than the headline, and it is the residual. The network is clustered relative to its own smoothed intensity at 250 m to 1.5 km in all fourteen years, and it was more clustered at 1 km in 2013 than the present network is inside the same footprint at the same density — a level difference, not a trend, since the fitted slope halves once 2013 is removed. Membership turns over continuously and does not accumulate, at a median replacement rate of 0.58, and that is the one result here not conditioned on the unit or the frame. Size and elapsed time cannot be separated at fourteen collinear points, so the exponent is the paper's one clean measurement and is attributed to neither.

Three consequences follow. For users of this network, the year of analysis is a conditioning variable that should be reported alongside any spatial result, and a registration rule that merges co-located channels is a reporting decision, not a preprocessing detail. That conditioning is not peculiar to this platform, and the reporting rule it implies is set out in the transferable rule: report the support of each year's pattern beside its window, the degree of registration co-location as a distribution, not a flag, and the rarefied series beside the unrarefied one. For those building longitudinal urban sensing datasets, births and last appearances are more informative than a net instrument count. And a frozen snapshot used as an execution target should carry a provenance statement, because the snapshot audited here covers 64.9% of the instruments that ever reported and omits vehicle counters, at an odds ratio of 2.55 when the comparison is conditioned on the snapshot's own year.

ACKNOWLEDGMENTS

The Newcastle Urban Observatory at Newcastle University is credited for maintaining an open, long-running municipal data platform; a census of this kind is only possible because the data are published without restriction.

ADDITIONAL INFORMATION AND DECLARATIONS

Funding

This research work was funded by Umm Al-Qura University, Saudi Arabia under grant number: 26UQU4350624GSSR07.

Grant Disclosures

The following grant information was disclosed by the author: Umm Al-Qura University (Deanship of Postgraduate Studies and Scientific Research), 26UQU4350624GSSR07.

Competing Interests

The authors prepared the benchmarking artifact whose provenance this paper audits, so the authors have a non-financial interest in the audit it receives; the audit is disclosed both in that section and here so that a reader can weight it. Apart from that, the authors declare no competing interests. The Newcastle Urban Observatory data the artifact derives from are independently published by Newcastle University (b).

Author Contributions

Ammar Arbaaeen: Funding acquisition, Writing – review and editing. Wael Alnahari: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review and editing, Visualization, Project administration. Tawfiq Hasanin: Validation, Writing – review and editing. Ammar Alsaig: Writing – review and editing. Wasem Alnahri: Conceptualization, Validation, Writing – review and editing.

Data Availability

The source data are the Newcastle Urban Observatory collection, openly archived by Newcastle University under CC BY 4.0 (27). The derived census, the fitted models, the snapshot audit and every analysis script behind the numbers in this paper are openly archived under CC BY 4.0 at `https://doi.org/10.5281/zenodo.23099939`, which always resolves to the current version, and are also provided as Supplemental Information. The deposit also carries the benchmark snapshot itself, the one-week instrument catalogue the snapshot audit is computed from, so that audit can be re-run from the deposit alone. That catalogue is a list of instruments — identity, position, broker and variable names — and every measured quantity in it has been removed; no temperature, concentration or count is redistributed, and none is read by any analysis. The deposit contains no observatory observations and no credential: what is deposited is our own work *about* the collection, together with the record of which instruments it was computed over. Related work on query performance over this platform is documented separately (Alnahari et al.).

Supplemental Information

Supplemental Information S1. Supporting material: the annual census and fitted models, the snapshot audit, and the analysis scripts that produce every number and figure in this paper.

Notes on contributors

Ammar Arbaaeen and **Ammar Alsaig** are in the Department of Computer Science, College of Computer and Information Systems, Umm Al-Qura University, Saudi Arabia. **Wael Alnahari** is in the Department of Information Systems, Faculty of Computing and Information Technology, King Abdulaziz University, Saudi Arabia. **Tawfiq Hasanin** is in the Department of Information Systems, King Abdulaziz University, Saudi Arabia. **Wasem Alnahri** is an independent researcher, Saudi Arabia.

Generative AI Use Disclosure Statement

Tool and version. Tool: GitHub Copilot, version 0.69.0. **How it was used.** Drafting and editing LaTeX and typesetting support. **Reason for use.** Assistance with manuscript preparation and language. It was not used to compute, alter or select any result; every number in the manuscript is read from a committed results file produced by the analysis scripts in the deposit. **Originality and accuracy.** The authors confirm the originality and accuracy of the content. **Terms of use.** The authors have checked the terms of use for this tool and confirm its use is suitable for publication. **Responsibility.** The authors take full responsibility for the integrity of the whole content, including the accuracy of all references.

REFERENCES

[Alegria et al.] Alegria, H., Dixon, B., Johns, R., Colarusso, A., and Hancock, M. Using a smart city framework to advance justice: applying an equity lens to compare air quality in neighbourhoods of different socioeconomic profiles. *Local Environment*, (10):1265–1286.

[Alnahari et al.] Alnahari, W., Hasanin, T., and Jambi, S. Complexity-tiered text-to-SQL performance of large language models in smart-city decision support. *IEEE Access*, pages 153491–153504.

[Altarrazi et al.] Altarrazi, S., Jha, D. N., Szydlo, T., and Ranjan, R. Data quality detector: Automating data quality detection in smart city environment. *2025 IEEE Symposium on Computers and Communications (ISCC)*, pages 1–6.

[Anthony] Anthony, B. Enabling seamless interoperability of digital systems in smart cities using API: A systematic literature review. *Journal of Urban Technology*, (4-5):123–156.

[Avram] Avram, S. Advancing urban analytics: GeoAI applications in spatial decision-making and sustainable cities. *Urban Science*, (3):148.

[Ayati et al.] Ayati, A., Naji, H. R., Hashemi, M. M., and Saffar, M. Optimizing location allocation in urban management: A brief review. In *2025 29th International Computer Conference, Computer Society of Iran (CSICC)*, pages 1–7.

[Baddeley and Turner] Baddeley, A. and Turner, R. spatstat: An R package for analyzing spatial point patterns. *Journal of Statistical Software*, (6).

[Barabesi et al.] Barabesi, L., Crescenzi, F., and Mori, L. Theil index decomposition: the influence function approach. *Statistical Papers*, (4):85.

[Barke] Barke, M. Morphogenesis and urban change in Newcastle upon Tyne: Context and process. In *The Urban Book Series*, pages 9–30. Springer Nature Switzerland.

[Barns] Barns, S. Smart cities and urban data platforms: Designing interfaces for smart governance. *City, Culture and Society*, pages 5–12.

[Benjamini and Hochberg] Benjamini, Y. and Hochberg, Y. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, (1):289–300.

[Cape] Cape, W. H. An overview of concept and practice—the urban observatory. *Midwest Review of Public Administration*, (2):121–123.

[Chen et al.] Chen, L., Grimstead, I., Bell, D., Karanka, J., Dimond, L., James, P., Smith, L., and Edwardes, A. Estimating vehicle and pedestrian activity from town and city traffic cameras. *Sensors*, (13):4564.

[Clark and Evans] Clark, P. J. and Evans, F. C. Distance to nearest neighbor as a measure of spatial relationships in populations. *Ecology*, (4):445–453.

[Davern et al.] Davern, M., Both, A., Murray, K., Roberts, R., and Norzahari, F. Liveability research creating real world impact: connecting urban planning and public health through the Australian Urban Observatory. *Cities & Health*, (5):765–778.

[Derbali et al.] Derbali, A., Dwidar, S., Abdelsattar, A., Sirror, H., and Abdelgawad, D. A review of the role of urban observatories in manipulating big data and ensuring sustainable urban governance in Saudi Arabia & Middle Eastern countries. In *2025 Eighth International Women in Data Science Conference at Prince Sultan University (WiDS PSU)*, pages 31–36.

[Donnelly] Donnelly, K. P. Simulations to determine the variance and edge effect of total nearest-neighbour distance. In Hodder, I., editor, *Simulation Studies in Archaeology*, pages 91–95. Cambridge University Press, Cambridge.

[Efron] Efron, B. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, (1).

[Enlund et al.] Enlund, D., Harrison, K., Ringdahl, R., Börütecene, A., Löwgren, J., and Angelakis, V. The role of sensors in the production of smart city spaces. *Big Data & Society*, (2):20539517221110218.

[Farhadi et al.] Farhadi, F., Palacin, R., and Blythe, P. Machine learning for transport policy interventions on air quality. *IEEE Access*, pages 43759–43777.

[Fernandes Barata et al.] Fernandes Barata, A., Harland, R. G., Kim, M., Larimian, T., Natapov, A., Palaiologou, F., and Weightman, J. Learning from the participatory practices of urban observatories to co-create a new town observatory in loughborough, UK. *Journal of Urban Affairs*, (5):1527–1543.

[Getis and Ord] Getis, A. and Ord, J. K. The analysis of spatial association by use of distance statistics. *Geographical Analysis*, (3):189–206.

[Gini] Gini, C. Measurement of inequality of incomes. *The Economic Journal*, (121):124–126.

[Hamed et al.] Hamed, N., Rana, O., Orozco-terWengel, P., Goossens, B., and Perera, C. A comparison of open data observatories. *Journal of Data and Information Quality*, (1):1–28.

[Huang et al.] Huang, X., Wang, S., Yang, D., Hu, T., Chen, M., Zhang, M., Zhang, G., Biljecki, F., Lu, T., Zou, L., Wu, C. Y. H., Park, Y. M., Li, X., Liu, Y., Fan, H., Mitchell, J., Li, Z., and Hohl, A. Crowdsourcing geospatial data for earth and human observations: A review. *Journal of Remote Sensing*,

page 0105.

[26] James, P., Jonczyk, J., Smith, L., Harris, N., Komar, T., Bell, D., and Ranjan, R. Realizing smart city infrastructure at scale, in the wild: A case study. *Frontiers in Sustainable Cities*, page 767942.

[27] James, P., Smith, L., Jonczyk, J., Harris, N., Komar, T., Puussaar, A., Clement, M., and Dawson, R. J. Urban observatory data newcastle. Open data collection, CC BY 4.0. Monthly and yearly CSV bundles, 2013–2026; accessed 29 September 2026. The unversioned DOI is the concept DOI and resolves to the current version.

[Jamshed et al.] Jamshed, M. A., Ali, K., Abbasi, Q. H., Imran, M. A., and Ur-Rehman, M. Challenges, applications, and future of wireless sensors in internet of things: A review. *IEEE Sensors Journal*, (6):5482–5494.

[Jiang and Ma] Jiang, F. and Ma, J. Environmental justice in the 15-minute city: Assessing air pollution exposure inequalities through machine learning and spatial network analysis. *Smart Cities*, (2):53.

[Karasaki et al.] Karasaki, S., Berberian, A. G., Zewdie, H. Y., Cushing, L. J., and VoPham, T. Geospatial approaches for environmental justice: A critical review. *Annual Review of Public Health*, (1):19–39.

[Keast et al.] Keast, L., Bramwell, L., Maji, K., Rankin, J., and Namdeo, A. Air quality outside schools in Newcastle upon Tyne, UK: An investigation into NO2 and PM concentrations and PM respiratory deposition. *Atmosphere*, (2):172.

[Kiefer and Vogelsang] Kiefer, N. M. and Vogelsang, T. J. A new asymptotic theory for heteroskedasticity-autocorrelation robust tests. *Econometric Theory*, (6):1130–1164.

[Komar and James] Komar, T. and James, P. Spatio-temporal hierarchical feature engineering for forecasting of urban footfall. *Applied Sciences*, (7):3162.

[Kumar et al.] Kumar, V., Gunner, S., Pregnolato, M., Tully, P., Georgalas, N., Oikonomou, G., Karatzas, S., and Tryfonas, T. Sense (and) the city: From internet of things sensors and open data platforms to urban observatories. *IET Smart Cities*, (4):291–311.

[Lakens et al.] Lakens, D., Scheel, A. M., and Isager, P. M. Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science*, (2):259–269.

[Li et al.] Li, J., Ranjan, R., Wang, Y., Huang, X., James, P., and Dustdar, S. The urban space information platform: Opportunities and challenges. *IEEE Internet Computing*, (3):66–75.

[Lin et al.] Lin, Y., Robinson, C., Yeap, Q. F., and Michael, H. Optimizing air pollution sensing for social and environmental justice. *Applied Geography*, page 103606.

[Longley] Longley, P. A. Smart data infrastructure for urban analytics, city science, planning and design. *Environment and Planning B: Urban Analytics and City Science*, (5):1041–1044.

[Malaker and Meng] Malaker, T. and Meng, Q. Urban disparity analytics using GIS: A systematic review. *Sustainability*, (14):5956.

[Mann] Mann, H. B. Nonparametric tests against trend. *Econometrica*, (3):245–259.

[Moran] Moran, P. A. P. Notes on continuous stochastic phenomena. *Biometrika*, (1/2):17–23.

[Morris-Wiltshire et al.] Morris-Wiltshire, C., Barr, S., James, P., and Komar, T. Quality-aware AIoT-enabled wireless urban sensor networks using deep learning. *Sensors*, (18):5825.

[Nazeeya Anjum et al.] Nazeeya Anjum, N., Sumathi, S., Dheena Dhayalan, R., Rampavithran, R., Jose Anand, A., and Sathiyanarayanan, M. Smart sensors for smart city environmental monitoring systems. In *Digital Twin, Blockchain, and Sensor Networks in the Healthy and Mobile City*, pages 17–33. Elsevier.

[Newey and West] Newey, W. K. and West, K. D. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, (3):703–708.

[Osamy et al.] Osamy, W., Khedr, A. M., Salim, A., Ali, A. I. A., and El-Sawy, A. A. Coverage, deployment and localization challenges in wireless sensor networks based on artificial intelligence techniques: A review. *IEEE Access*, pages 30232–30257.

[Páez and Scott] Páez, A. and Scott, D. M. Spatial statistics for urban analysis: A review of techniques with examples. *GeoJournal*, (1):53–67.

[Prais and Winsten] Prais, S. J. and Winsten, C. B. Trend estimators and serial correlation. Technical Report 383, Cowles Commission for Research in Economics, University of Chicago.

[Puussaar et al.] Puussaar, A., Montague, K., Peacock, S., Nappey, T., Anderson, R., Jonczyk, J., Wright, P., and James, P. SenseMyStreet: Sensor commissioning toolkit for communities. *Proceedings of the ACM on Human-Computer Interaction*, (CSCW2):1–26.

[Ripley] Ripley, B. D. Modelling spatial patterns. *Journal of the Royal Statistical Society Series B:*

Statistical Methodology, (2):172–192.

[Robinson and Franklin] Robinson, C. and Franklin, R. S. The sensor desert quandary: What does it mean (not) to count in the smart city? *Transactions of the Institute of British Geographers*, (2):238–254.

[Robinson et al.] Robinson, C., Franklin, R. S., and Roberts, J. Optimizing for equity: Sensor coverage, networks, and the responsive city. *Annals of the American Association of Geographers*, (8):2152–2173.

[Rusli et al.] Rusli, N., Ling, G. H. T., Mohd Hussain, M. H., Mohamed Salib, N. S., Abu Bakar, S. Z., and Othman, M. H. A review on worldwide urban observatory systems' data analytics themes: Lessons learned for Malaysia Urban Observatory (MUO). *Journal of Urban Management*, (3):231–254.

[Schuirmann] Schuirmann, D. J. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, (6):657–680.

[Sen] Sen, P. K. Estimates of the regression coefficient based on kendall's tau. *Journal of the American Statistical Association*, (324):1379–1389.

[Sheeja Rani et al.] Sheeja Rani, S., Aburukba, R., and El Fakih, K. Wireless sensor networks for urban development: A study of applications, challenges, and performance metrics. *Smart Cities*, (3):89.

[Theil] Theil, H. *Economics and Information Theory*. North-Holland, Amsterdam, The Netherlands. Studies in Mathematical and Managerial Economics, Vol. 7.

[Theodorakopoulos and Theodoropoulou] Theodorakopoulos, L. and Theodoropoulou, A. Big data analytics for geospatial decision-making in smart cities: A review of spatial data, GeoAI and urban digital twins. *ISPRS International Journal of Geo-Information*, (7):278.

[58] Wolf, K., Dawson, R. J., Mills, J. P., Blythe, P., Robson, C., and Morley, J. Assessing the impact of heavy rainfall on the Newcastle upon Tyne transport network using a geospatial data infrastructure. *Resilient Cities and Structures*, (2):24–41.

[59] Wolf, K., Stiles, J., Miller, H. J., Dawson, R. J., Mills, J. P., Blythe, P., and Morley, J. Building enduring smart city data platforms to provide urban management support: lessons learnt from UK urban observatories and the US Smart Columbus Operating System. *Frontiers in Sustainable Cities*.

[Wood] Wood, A. The myth of the "Brasília of the North": rhetoric and reality in newcastle's urban transformation. *Urban Geography*, (1):88–107.

[Xia et al.] Xia, N., Bian, J., Wang, C., and Wang, M.-J.-S. Measurement-dependent asymmetry in chinese urban carbon-emission inventories: Reference states, scale sensitivity, and cross-inventory concordance. *Symmetry*, (9):1578.

[Xu et al.] Xu, A., Biehl, J., and Lee, A. Getting it just right: Towards balanced utility, privacy, and equity in shared space sensing. *ACM Transactions on Internet of Things*, (2):1–26.

[Yu] Yu, D. Toward integrated urban observatories: Synthesizing remote and social sensing in urban science. *Remote Sensing*, (12):2041.

[Yu and Fang] Yu, D. and Fang, C. Urban remote sensing with spatial big data: A review and renewed perspective of urban studies in recent decades. *Remote Sensing*, (5):1307.

[Zeng et al.] Zeng, F., Pang, C., and Tang, H. Sensors on internet of things systems for the sustainable development of smart cities: A systematic literature review. *Sensors*, (7):2074.

[Zhai et al.] Zhai, W., Bai, X., Shi, Y., Tang, J., and Shi, Y. Low-cost sensor desert and equity across US cities. *Environment and Planning B: Urban Analytics and City Science*, (4):897–914.

[Zied Abozied et al.] Zied Abozied, E., Robinson, C., Franklin, R., Court, K., and Roberts, J. A spatial decision support framework for equitable sensor network distribution in the smart city. *The Geographical Journal*, (1):e12605.